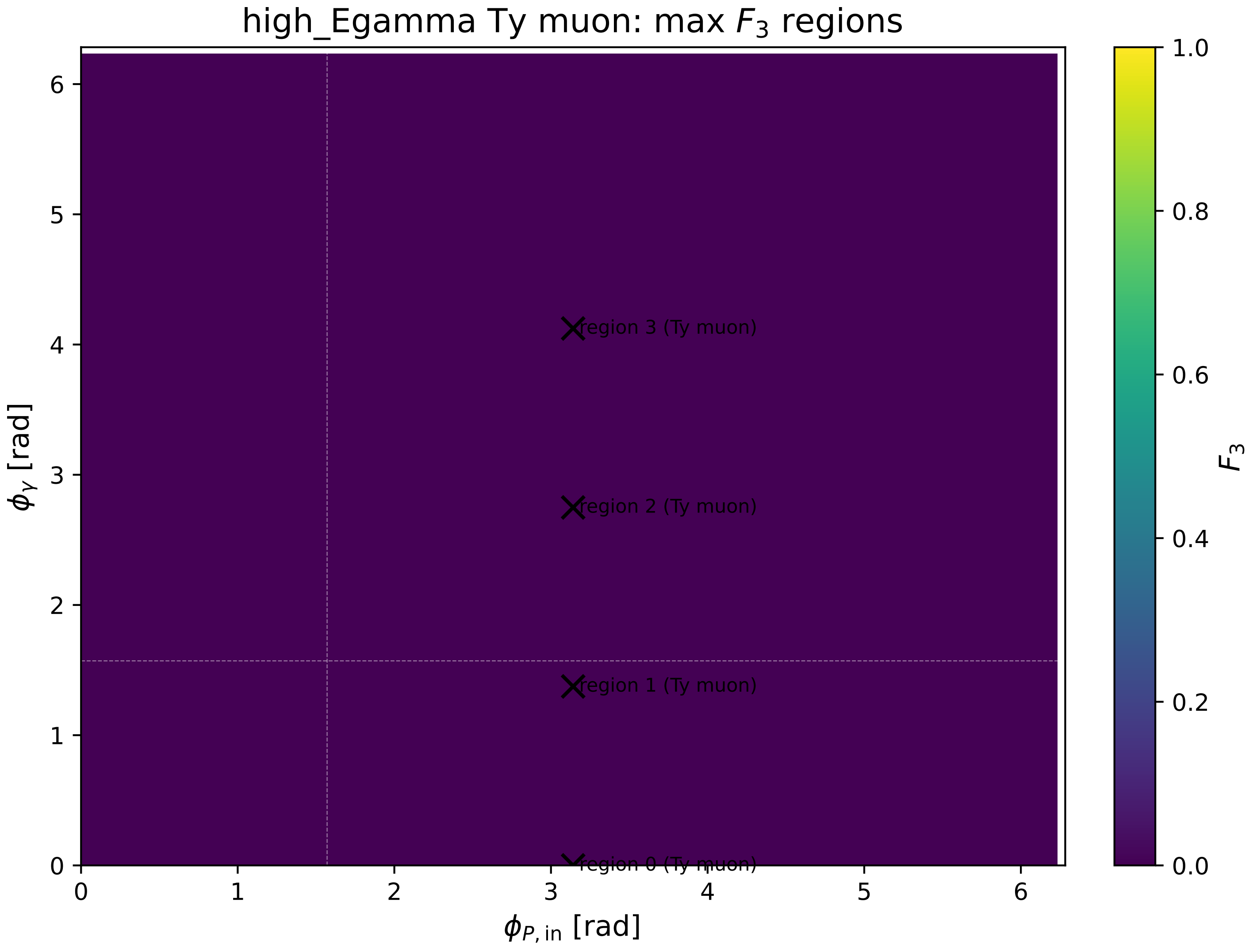
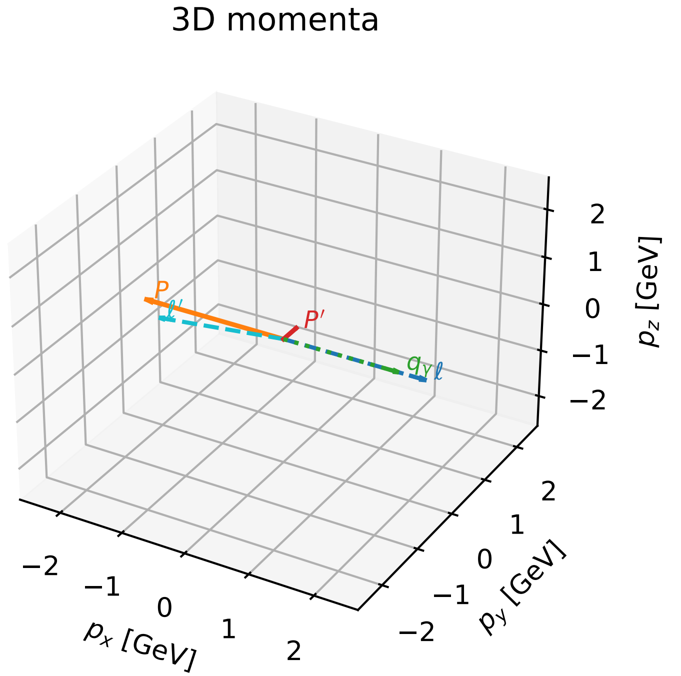

high_Egamma Ty muon: max $F_3$ regions
region 3 (Ty muon)
region 2 (Ty muon)
region 1 (Ty muon)
region 0 (Ty muon)
$\phi_\gamma$ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$F_3$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

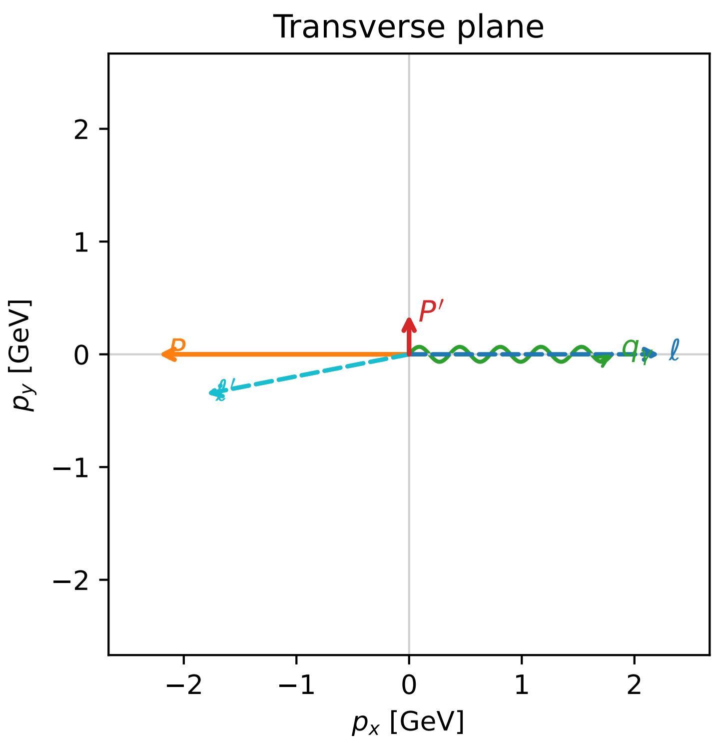
# Ty muon characteristic max $F_3$ configuration

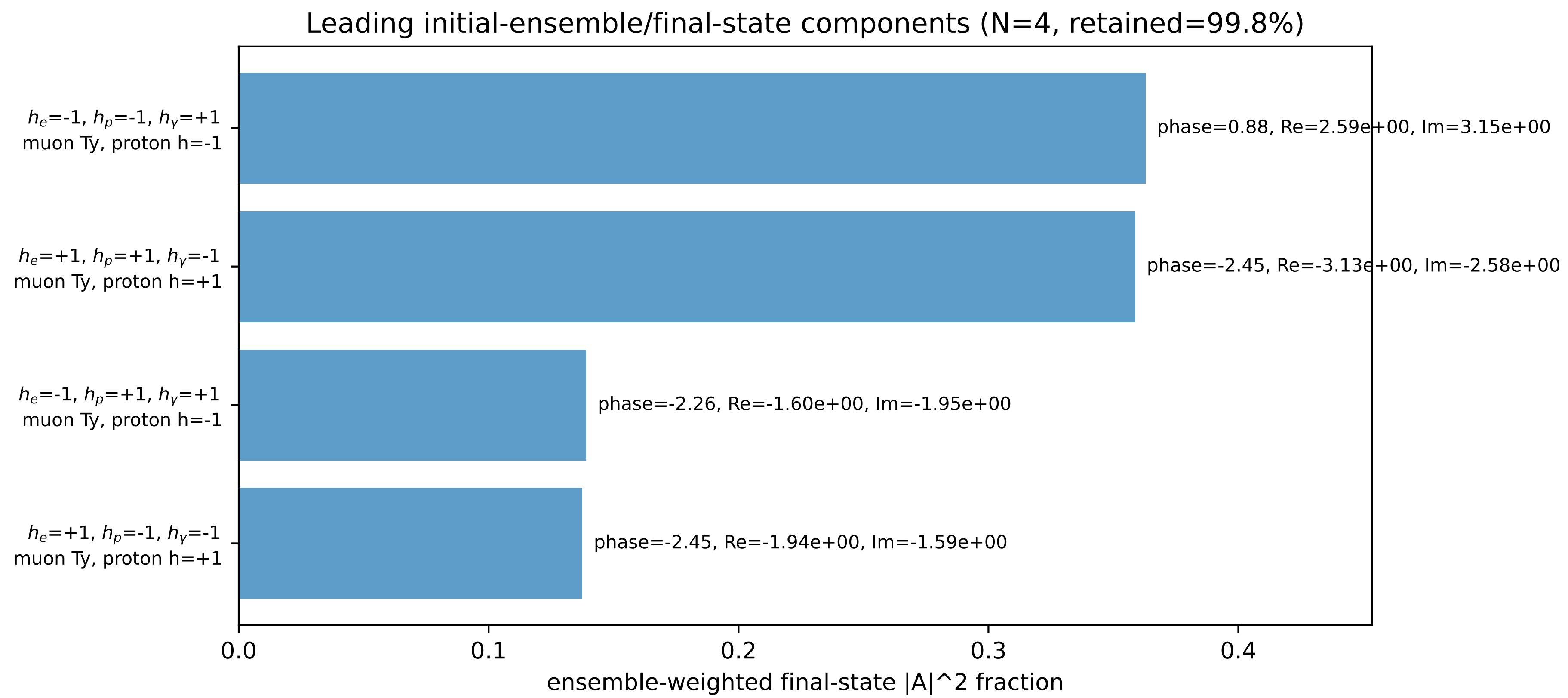



f3_high_egamma_ty_electron_region_0 F_3=0
region: high_Egamma_Ty_electron_region_0
outgoing-state purity: 0.500016
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\mu}, h_p)$

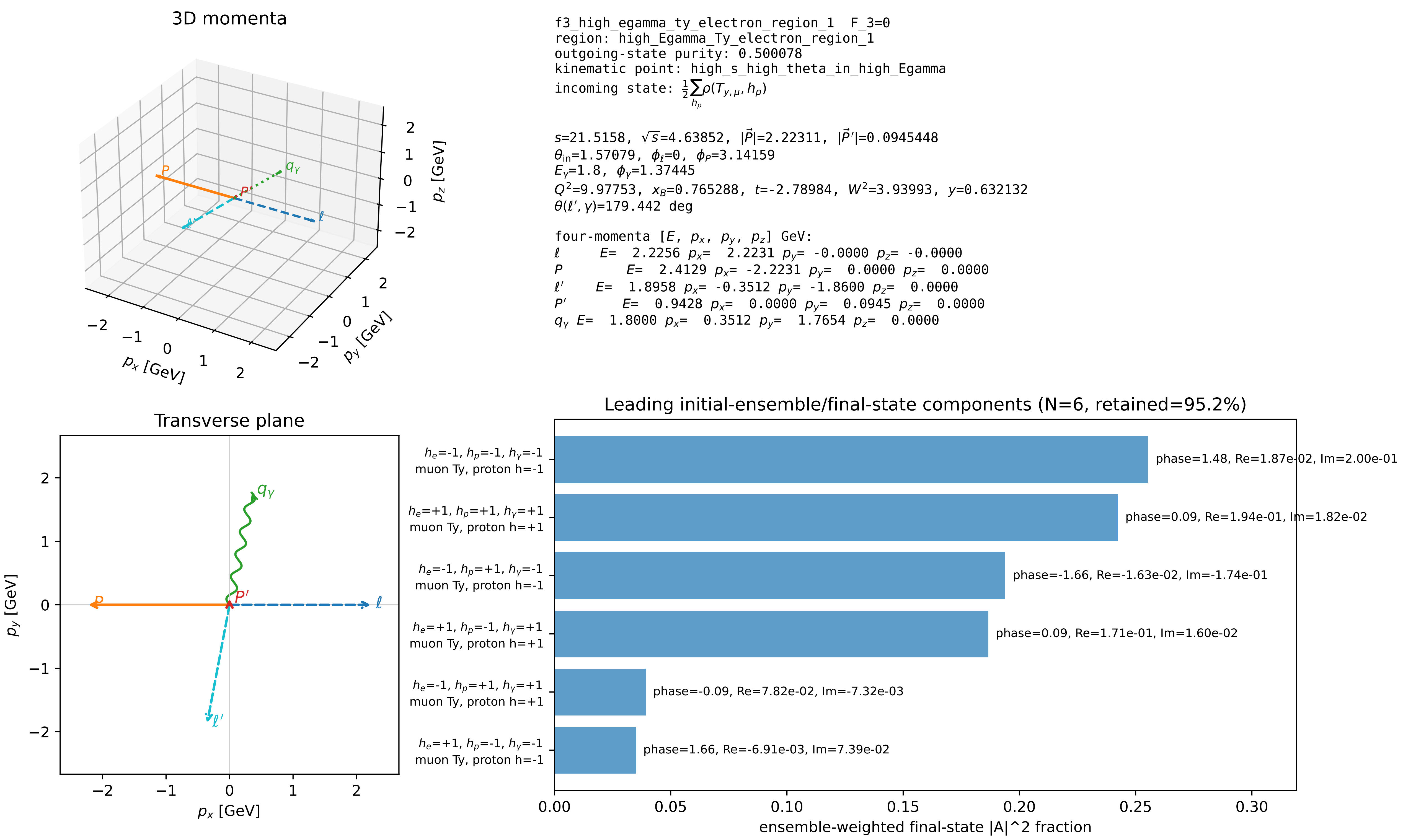
$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.351095
$\theta_{\text{in}}$=1.57079, $\phi_\ell$=0, $\phi_P$=3.14159
$E_\gamma$=1.8, $\phi_\gamma$=0
$Q^2$=16.1576, $x_B$=0.81756, $t$=-3.07361, $W^2$=4.48544, $y$=0.958225
$\theta(\ell', \gamma)$=168.963 deg

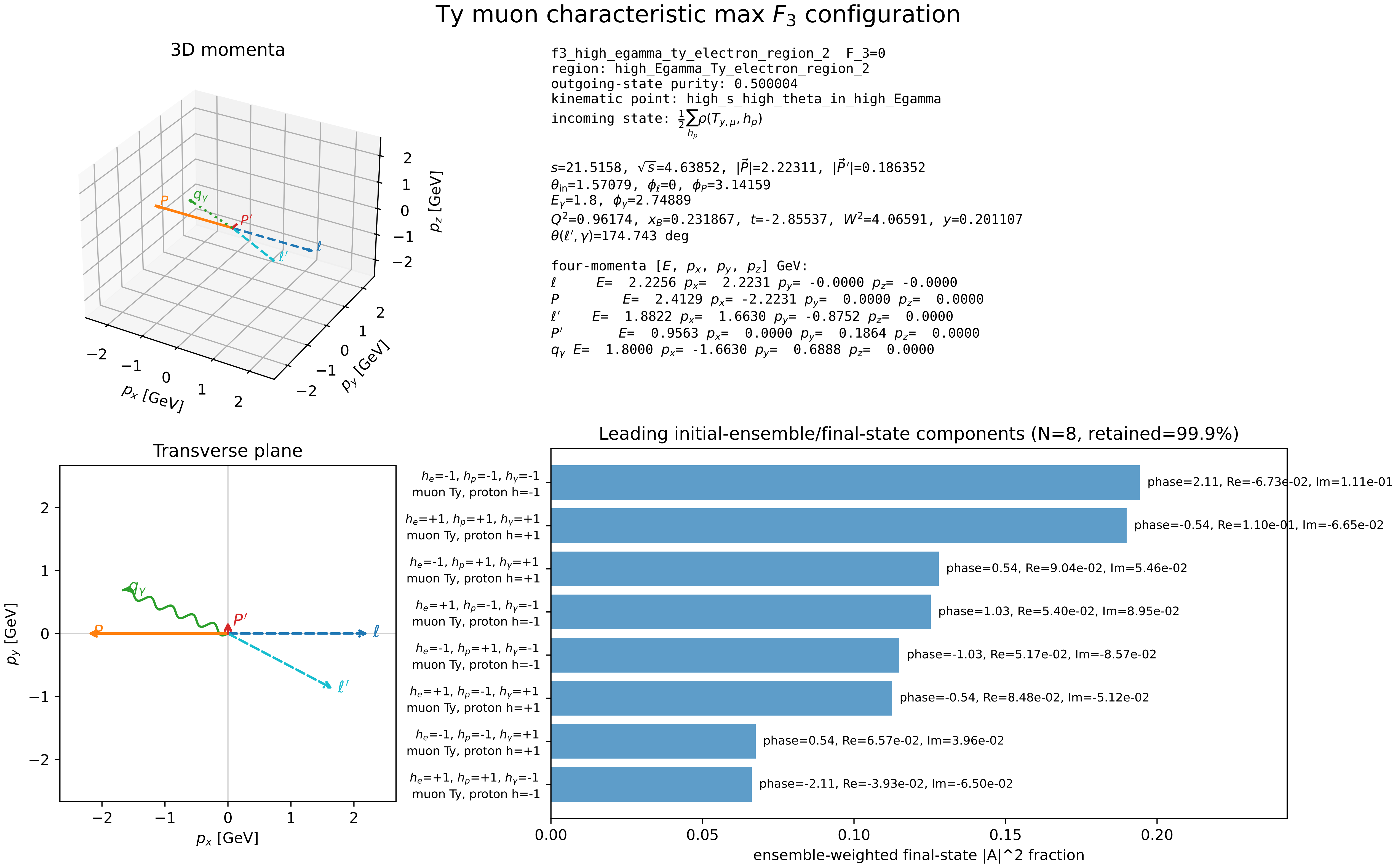
four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2256 $p_x$= 2.2231 $p_y$= -0.0000 $p_z$= -0.0000
$P$ $E$= 2.4129 $p_x$= -2.2231 $p_y$= 0.0000 $p_z$= 0.0000
$\ell'$ $E$= 1.8370 $p_x$= -1.8000 $p_y$= -0.3511 $p_z$= 0.0000
$P'$ $E$= 1.0016 $p_x$= 0.0000 $p_y$= 0.3511 $p_z$= 0.0000
$q_\gamma$ $E$= 1.8000 $p_x$= 1.8000 $p_y$= 0.0000 $p_z$= 0.0000



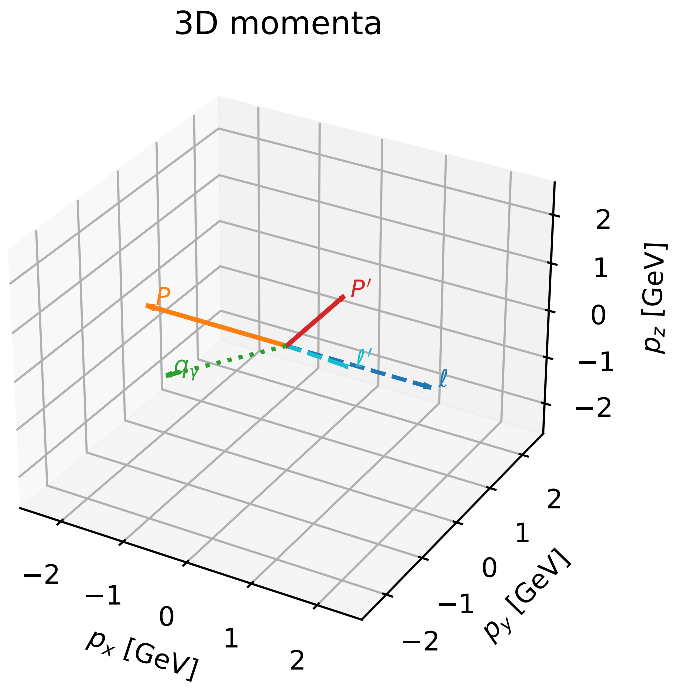



# Ty muon characteristic max $F_3$ configuration




Ty muon characteristic max $F_3$ configuration
3D momenta
$p_x$ [GeV]
$p_y$ [GeV]
$p_z$ [GeV]
$P$
$q_\gamma$
$P'$
$\ell$
$\ell'$
f3_high_egamma_ty_electron_region_2 F_3=0
region: high_Egamma_Ty_electron_region_2
outgoing-state purity: 0.500004
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\mu}, h_p)$
$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.186352
$\theta_{\mathrm{in}}$=1.57079, $\phi_\ell$=0, $\phi_P$=3.14159
$E_\gamma$=1.8, $\phi_\gamma$=2.74889
$Q^2$=0.96174, $x_B$=0.231867, $t$=-2.85537, $W^2$=4.06591, $y$=0.201107
$\theta(\ell', \gamma)$=174.743 deg
four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2256 $p_x$= 2.2231 $p_y$= -0.0000 $p_z$= -0.0000
$P$ $E$= 2.4129 $p_x$= -2.2231 $p_y$= 0.0000 $p_z$= 0.0000
$\ell'$ $E$= 1.8822 $p_x$= 1.6630 $p_y$= -0.8752 $p_z$= 0.0000
$P'$ $E$= 0.9563 $p_x$= 0.0000 $p_y$= 0.1864 $p_z$= 0.0000
$q_\gamma$ $E$= 1.8000 $p_x$= -1.6630 $p_y$= 0.6888 $p_z$= 0.0000
Transverse plane
$p_x$ [GeV]
$p_y$ [GeV]
Leading initial-ensemble/final-state components (N=8, retained=99.9%)
$h_e$=-1, $h_p$=-1, $h_\gamma$=-1 muon Ty, proton h=-1 — phase=2.11, Re=-6.73e-02, Im=1.11e-01
$h_e$=+1, $h_p$=+1, $h_\gamma$=+1 muon Ty, proton h=+1 — phase=-0.54, Re=1.10e-01, Im=-6.65e-02
$h_e$=-1, $h_p$=+1, $h_\gamma$=+1 muon Ty, proton h=+1 — phase=0.54, Re=9.04e-02, Im=5.46e-02
$h_e$=+1, $h_p$=-1, $h_\gamma$=-1 muon Ty, proton h=-1 — phase=1.03, Re=5.40e-02, Im=8.95e-02
$h_e$=-1, $h_p$=+1, $h_\gamma$=-1 muon Ty, proton h=-1 — phase=-1.03, Re=5.17e-02, Im=-8.57e-02
$h_e$=+1, $h_p$=-1, $h_\gamma$=+1 muon Ty, proton h=+1 — phase=-0.54, Re=8.48e-02, Im=-5.12e-02
$h_e$=-1, $h_p$=-1, $h_\gamma$=+1 muon Ty, proton h=+1 — phase=0.54, Re=6.57e-02, Im=3.96e-02
$h_e$=+1, $h_p$=+1, $h_\gamma$=-1 muon Ty, proton h=-1 — phase=-2.11, Re=-3.93e-02, Im=-6.50e-02
ensemble-weighted final-state |A|^2 fraction

# Ty muon characteristic max $F_3$ configuration




```
f3_high_egamma_ty_electron_region_3  F_3=0
region: high_Egamma_Ty_electron_region_3
outgoing-state purity: 0.500091
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\mu}, h_p)$

$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.57149
$\theta_{\text{in}}$=1.57079, $\phi_\ell$=0, $\phi_P$=3.14159
$E_\gamma$=1.8, $\phi_\gamma$=4.12334
$Q^2$=0.019849, $x_B$=0.00175464, $t$=-7.0722, $W^2$=12.1723, $y$=0.548479
$\theta(\ell', \gamma)$=119.47 deg

four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2256 $p_x$= 2.2231 $p_y$= -0.0000 $p_z$= -0.0000
$P$ $E$= 2.4129 $p_x$= -2.2231 $p_y$= 0.0000 $p_z$= 0.0000
$\ell'$ $E$= 1.0084 $p_x$= 1.0000 $p_y$= -0.0748 $p_z$= 0.0000
$P'$ $E$= 1.8301 $p_x$= 0.0000 $p_y$= 1.5715 $p_z$= 0.0000
$q_\gamma$ $E$= 1.8000 $p_x$= -1.0000 $p_y$= -1.4966 $p_z$= 0.0000